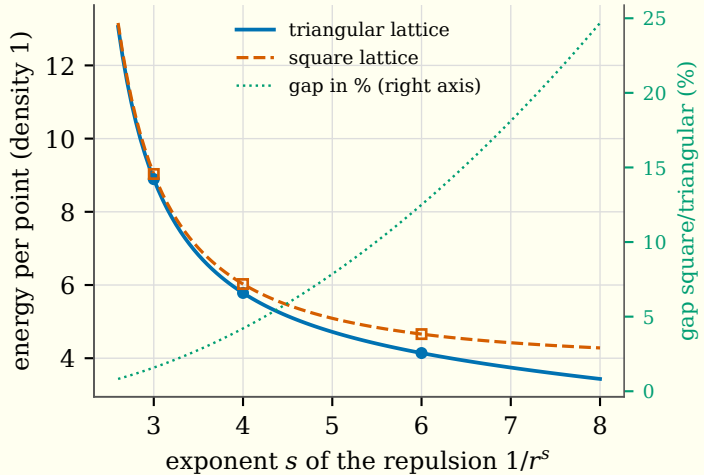
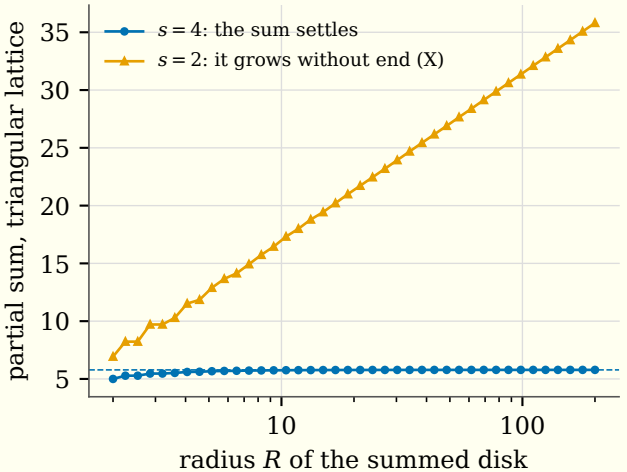

The triangular lattice is always lower
triangular lattice
square lattice
gap in % (right axis)
energy per point (density 1)
gap square/triangular (%)
exponent s of the repulsion $1/r^s$
12
10
8
6
4
25
20
15
10
5
0
3
4
5
6
7
8


Above $s = 2$, a finite minimum
$s = 4$: the sum settles
$s = 2$: it grows without end (X)
partial sum, triangular lattice
radius R of the summed disk
35
30
25
20
15
10
5
10
100